CS F320

Foundations of Data Science

Assignment 1 — Report

PROBLEMS

P1 Polynomial regression: closed form vs. the gradient-descent family

P2 Ridge regression as a MAP estimate

P3 L1-regularised regression via subgradient descent

P4 Bayesian regression: MAP/ridge equivalence and the effect of N

TEAM MEMBERS

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Shuffle seed (P1: SGD and mini-batch GD): **42** • 19 October 2026

Overview

All four problems were implemented in Python with NumPy and Matplotlib only, loading the provided CSV files directly. No data were generated or modified, and $\sin(2\pi x)$ is used for plotting only, never for fitting. Every number in this report is copied from the console output saved by the corresponding script (`results/P1_output.txt` to `results/P4_output.txt`).

Files: `p1_gd_family.py`, `p2_ridge_map.py`, `p3_lasso.py`, `p4_bayesian.py` (one script per problem), `common.py` (shared helpers and the seed, SEED = 42), `data/` (the six provided CSV files), `figs/` (all figures), `results/` (console outputs). See `README.md` for how to run everything.

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P1

Polynomial Regression — Closed Form vs the Gradient Descent Family

Method

Closed form: the weights for each degree d = 1–9 are computed with the SVD-based `np.linalg.lstsq`, never with an explicit inverse. For d = 9 the Vandermonde matrix is ill-conditioned, so gradient descent is run on the QR-reparametrised problem $E(\alpha) = 0.5\,\|Q\alpha - t\|^2$ with $\alpha = R\,w$, starting from $\alpha = 0$, and the weights are recovered with $w = R^{-1}\,\alpha$. Batch GD uses the full-sum gradient (not divided by N); SGD reshuffles each epoch and makes N = 10 single-sample updates; mini-batch GD (size 3, batches 3, 3, 3, 1) reshuffles each epoch and uses the averaged gradient.

Closed-form fits, d = 1–9

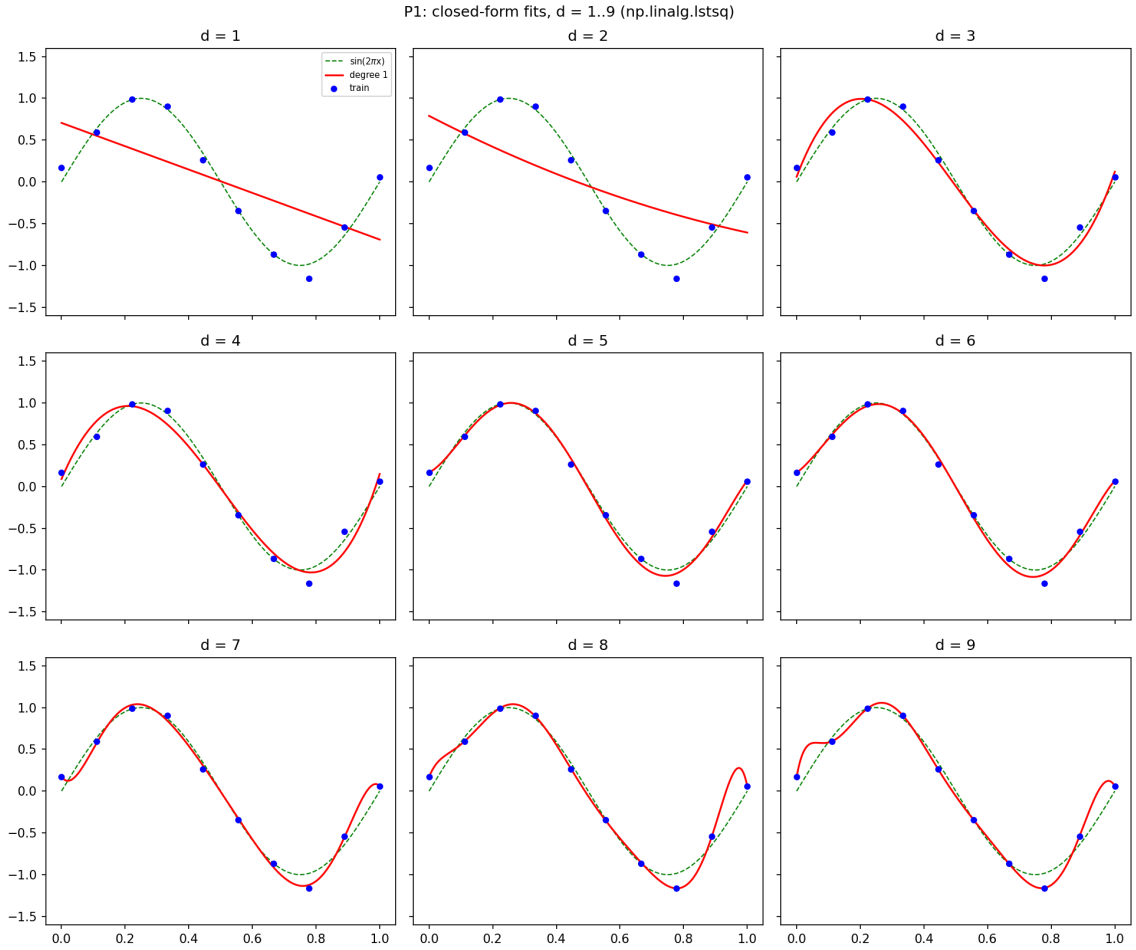


Figure 1. Closed-form polynomial fits for d = 1–9 against the training points and $\sin(2\pi x)$.

Training SSE of the closed-form fit for each degree

```
degree     train SSE E(w)
     1       1.386903e+00
     2       1.373940e+00
     3       8.443037e-02
     4       8.082228e-02
     5       1.655523e-02
     6       1.610836e-02
     7       6.274067e-03
     8       3.093784e-04
     9       6.247240e-23
```

Setup for d = 9

```
N = 10,  shuffle seed = 42
cond(Phi)            = 1.519e+07   (ill-conditioned)
cond(Q)              = 1.000e+00   (perfectly conditioned)
||Q^T Q - I||_F      = 1.11e-15
max |Q Q^T - I|      = 4.44e-16   (rows orthonormal too)
||w_closed||_2       = 1.0918e+05
w_closed:
   w[0] = +1.6905e-01
   w[1] = +2.1269e+01
   w[2] = -4.1353e+02
   w[3] = +3.7880e+03
   w[4] = -1.7589e+04
   w[5] = +4.5606e+04
   w[6] = -6.9409e+04
   w[7] = +6.1610e+04
   w[8] = -2.9469e+04
   w[9] = +5.8558e+03
```

Loss curves (η = 0.1)

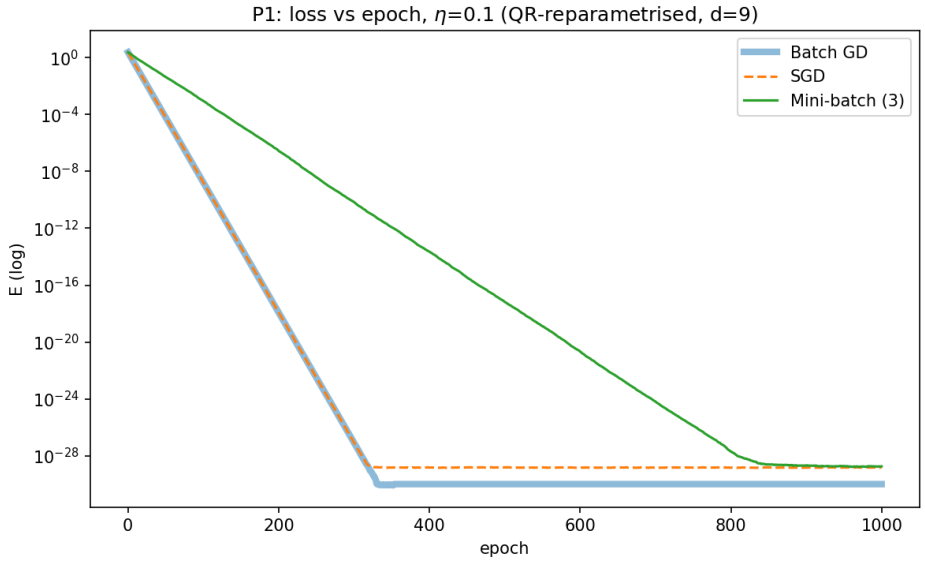


Figure 2. E vs epoch for Batch, SGD and mini-batch GD (log scale), η = 0.1.

3. LOSS CURVES (η = 0.1, QR basis, E = SSE)

```
 epoch          Batch            SGD     Mini-batch
     0     2.3807e+00     2.3807e+00     2.3807e+00
     1     1.9284e+00     1.9284e+00     2.2064e+00
     5     8.3010e-01     8.3010e-01     1.5479e+00
    10     2.8944e-01     2.8944e-01     1.0296e+00
    50     6.3234e-05     6.3234e-05     4.3660e-02
   100     1.6796e-09     1.6796e-09     9.0274e-04
   200     1.1850e-18     1.1850e-18     2.7307e-07
   500     1.0250e-30     1.5236e-29     6.6205e-18
  1000     1.0250e-30     1.4841e-29     1.8169e-29
  2000     1.0250e-30     1.4762e-29     1.5163e-29

max |E_batch - E_sgd| over the first 50 epochs = 4.44e-16
measured E shrink per epoch : Batch = 0.8100  (theory (1-eta)^2     = 0.8100)
                              Mini  = 0.9248  (theory (1-eta/3)^2   = 0.9344, slightly lower because the singleton batch is faster)
```

Learning-rate study

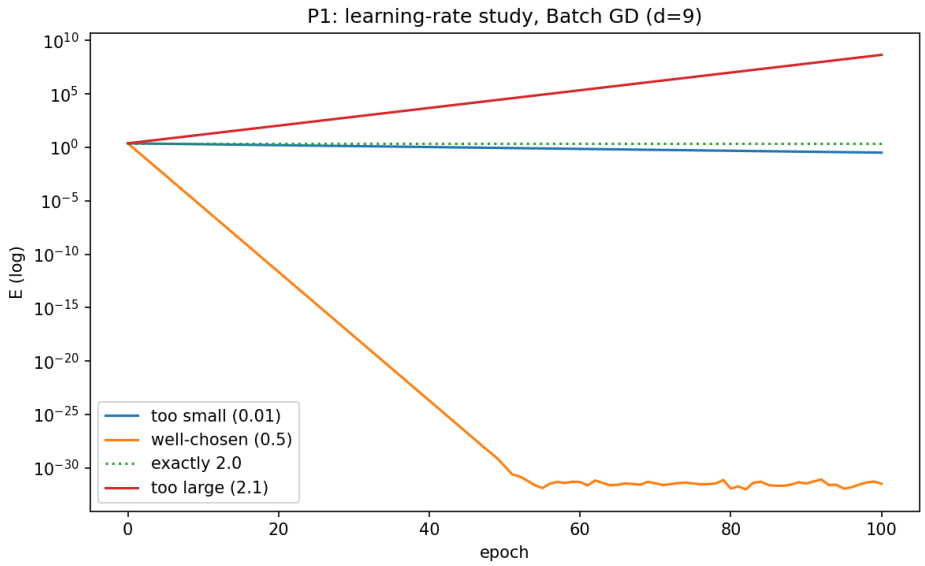


Figure 3. Batch GD for too small (0.01), well-chosen (0.5), too large (2.1) and exactly 2.0.

4. LEARNING-RATE STUDY (Batch GD, QR basis, 100 epochs)

```
   eta   E after 100 epochs   regime
  0.01           3.1896e-01   too small: slow
  0.10           1.6796e-09   good
  0.50           3.2741e-32   good: fast
  1.00           8.6498e-32   good: fastest (one step)
  2.00           2.3807e+00   boundary: oscillates, never converges
  2.10           4.5211e+08   too large: diverges

eta = 2.1 after 100 epochs: E = 4.5211e+08
   w = (original basis, w = R^-1 alpha)
   w[0] = -2.3295e+03
   w[1] = -2.9307e+05
   w[2] = +5.6983e+06
   w[3] = -5.2197e+07
   w[4] = +2.4236e+08
   w[5] = -6.2843e+08
   w[6] = +9.5644e+08
   w[7] = -8.4896e+08
   w[8] = +4.0607e+08
   w[9] = -8.0691e+07

eta = 2.0: first E values [2.3807 2.3807 2.3807 2.3807] -> flips between alpha=0 and alpha=2 Q^T t, same loss each time
eta = 0.01 is slow but converges: E at epochs 100/500/1000/2000 = 3.190e-01, 1.028e-04, 4.437e-09, 8.270e-18
```

Verification against the closed form

5. VERIFICATION vs CLOSED FORM (2000 epochs, η = 0.1)

```
||w_closed|| = 1.0918e+05
method          ||w_GD - w_cf||_2   relative error
Batch                 7.7664e-06       7.1135e-11
SGD                   7.7665e-06       7.1136e-11
Mini-batch            7.7661e-06       7.1132e-11

The absolute error looks large-ish only because ||w|| ~ 1e5; the relative error ~1e-11 is rounding level.
```

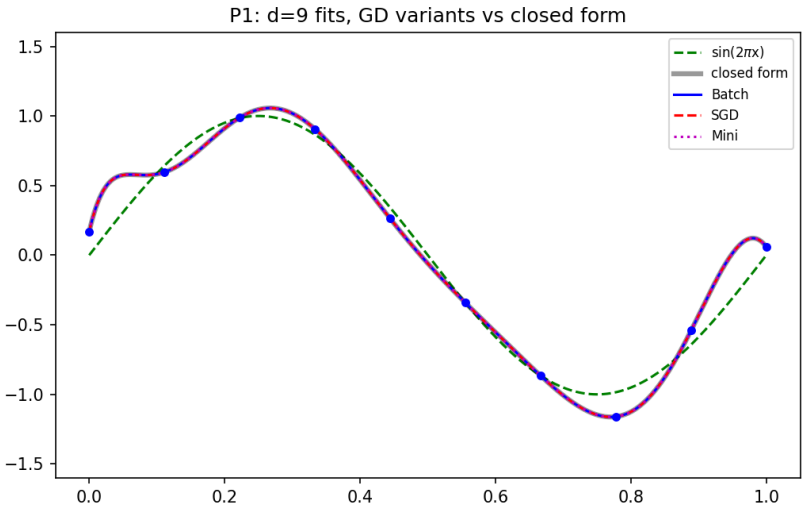


Figure 4. The d = 9 fit obtained by each GD variant coincides with the closed-form fit.

Discussion

Q is square (10×10) and orthogonal, so its ROWS are orthonormal too. An SGD update for sample n moves α only along q_n, so it leaves every other residual untouched, and each residual shrinks by exactly $(1 - \eta)$ per epoch whatever the shuffle order: identical to Batch GD, hence the curves coincide. Mini-batch GD averages the gradient, dividing the step by the batch size, so residuals in 3-sample batches shrink only by $(1-\eta/3)$ per epoch (only the singleton batch uses the full η). The slowest modes dominate E, so it converges more slowly than both.

P2

Ridge Regression as a MAP Estimate

Method

A zero-mean Gaussian prior on the weights with a Gaussian likelihood gives a MAP estimate equal to the ridge solution with $\lambda = \alpha/\beta$. The closed form solves $(\Phi^\top \Phi + \lambda\, \mathrm{I})\ \mathrm{w} = \Phi^\top \mathrm{t}$ as a linear system (no explicit inverse) in the $\mathrm{d} = 9$ monomial basis, for every λ on `np.logspace(-10, 2, 49)`. Gradient descent is run directly in the original basis from $\mathrm{w} = 0$ with $\eta = 0.04$ at $\lambda = 10^{-2}$, until the relative error to the closed form is below 10^{-6}.

Closed-form sweep

lambda	train RMSE	test RMSE	||w||_2
1.000e-10	0.0165	0.1605	3.979e+03
1.000e-09	0.0245	0.1461	1.341e+03
1.000e-08	0.0320	0.1297	7.962e+02
1.000e-07	0.0514	0.1151	1.944e+02
1.000e-06	0.0580	0.1172	4.672e+01
1.000e-05	0.0611	0.1222	2.658e+01
1.000e-04	0.0676	0.1199	2.187e+01
1.000e-03	0.1253	0.1511	1.487e+01
1.000e-02	0.2659	0.2660	5.108e+00
1.000e-01	0.3558	0.3513	2.254e+00
1.000e+00	0.5154	0.5158	8.321e-01
1.000e+01	0.6436	0.6595	1.568e-01
1.000e+02	0.6815	0.7058	2.359e-02

(table shows every 4th grid point; full grid is used for all results below)

Gradient descent vs closed form at λ = 10^{-2}

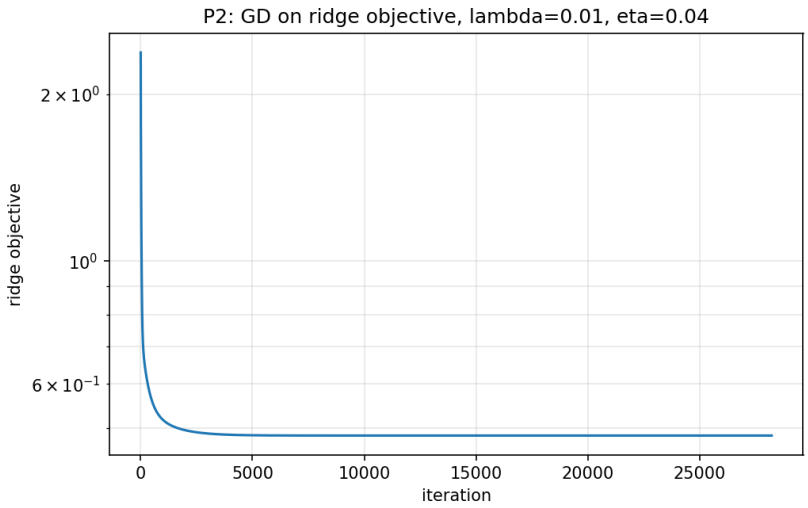


Figure 5. Ridge objective vs iteration for gradient descent at λ = 0.01 (η = 0.04).

2. GRADIENT DESCENT vs CLOSED FORM at λ = 0.01 (original basis, η = 0.04)

```
lambda_max(Phi^T Phi + lam I) = 20.381
stability needs eta < 2/lambda_max = 0.0981  -> eta = 0.04 is SAFE

GD reached relative error < 1e-6 after 28227 iterations
final relative error ||w_GD - w_cf|| / ||w_cf|| = 9.997e-07

 iteration    ridge objective   relative error
         0       2.380690e+00        1.000e+00
         1       2.173812e+00        9.944e-01
        10       1.646042e+00        9.548e-01
       100       7.109328e-01        8.073e-01
      1000       5.164352e-01        4.312e-01
      5000       4.846395e-01        6.274e-02
     10000       4.840330e-01        5.791e-03
     20000       4.840278e-01        4.955e-05
     28227       4.840278e-01        9.997e-07

cond(Phi^T Phi)           = 2.307e+14
cond(Phi^T Phi + lam I)   = 2.038e+03   (reduced by a factor of 1.1e+11)
The penalty adds lambda to every eigenvalue, lifting the near-zero ones, so the condition number collapses.
```

RMSE sweep and choice of λ^*

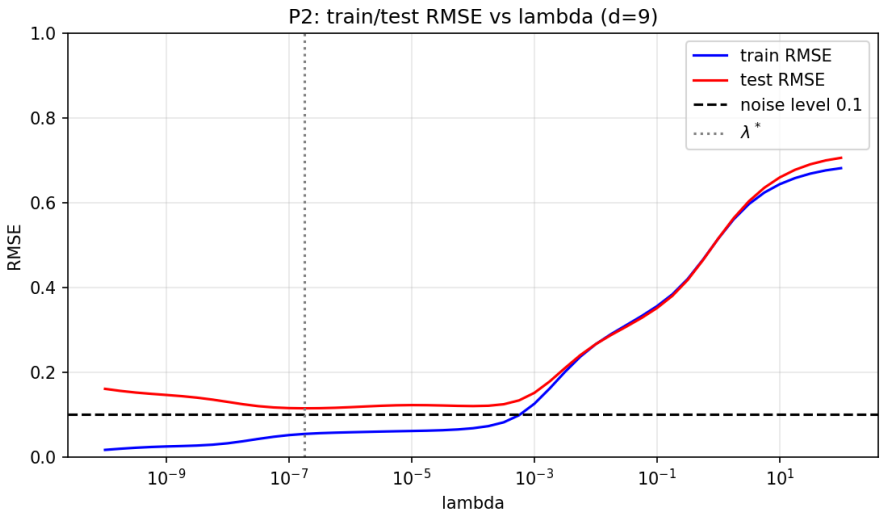


Figure 6. Train and test RMSE vs λ; the dashed line is the noise level 0.1.

3. RMSE SWEEP AND CHOICE OF λ^*

```
lambda* = argmin test RMSE      = 1.7783e-07
test RMSE at lambda*            = 0.1146   (noise level = 0.1)
train RMSE at lambda*           = 0.0541
interval with test RMSE <= 1.05 x min : [3.162e-08, 1.000e-04]   (9 grid points, contiguous = False)
min test RMSE = 0.1146; threshold = 0.1203
grid points inside the 1.05 x min band: 3.16e-08, 5.62e-08, 1.00e-07, 1.78e-07, 3.16e-07, 5.62e-07, 1.00e-06, 1.78e-06, 1.00e-04
over the whole span, test RMSE ranges only from 0.1146 to 0.1222
```

Fitted curves

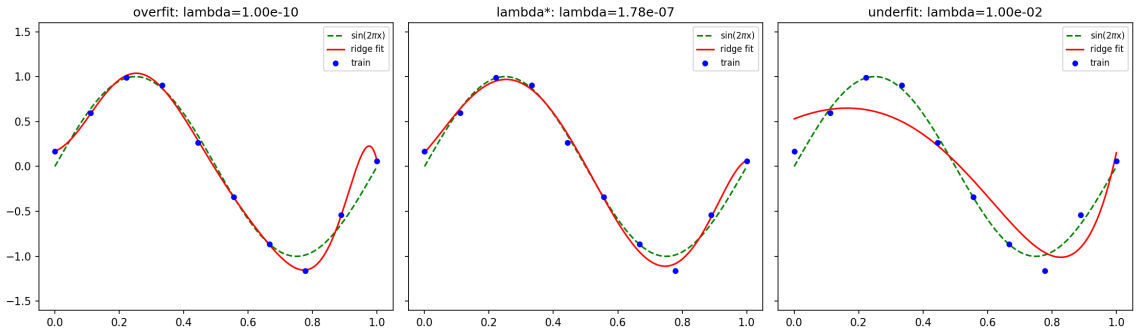


Figure 7. Overfit ($\lambda = 10^{-10}$), λ^* and underfit ($\lambda = 10^{-2}$) fits.

4. THREE FITS: overfit / λ^* / underfit

case	lambda	||w||_2	train RMSE	test RMSE
overfit	1.000e-10	3.9792e+03	0.0165	0.1605
lambda*	1.778e-07	1.2449e+02	0.0541	0.1146
underfit	1.000e-02	5.1079e+00	0.2659	0.2660

Discussion

Ridge is the MAP estimate under a zero-mean Gaussian prior, with $\lambda = \alpha/\beta$. The test-RMSE minimum is FLAT: 9 grid points lie within 5% of the best (0.115), spanning λ from 3.2×10^{-8} to 10^{-4} (about 3.5 decades), and over that whole span test RMSE only moves between 0.115 and 0.122. Below the interval the fit overfits (train RMSE $\to$ 0, huge weights, wiggles between the points). Above it the penalty dominates and the fit underfits (weights shrink towards 0, the curve is too smooth, and both train and test RMSE rise).

P3

L1-Regularised Regression via Subgradient Descent

Method

The model is degree 9 in the shifted Legendre basis, $Phi_{nj} = P_j(2x_n - 1)$, which spans the same space as the monomial basis but is far better conditioned. The Lasso objective $\mathrm{E(w)} = 0.5\ \|\Phi w - t\|^2 + \lambda\ \|w\|_1$ is minimised by the subgradient update $\mathrm{w} \leftarrow \mathrm{w} - \eta\ (\Phi^\top(\Phi w - t) + \lambda\ \mathrm{sign(w)})$ with $\eta = 0.05$, starting from $\mathrm{w} = 0$, for 20,000 iterations at each λ on `np.logspace(-4, 0, 21)`. A coefficient is called approximately zero when $|w_j| < 2\ \eta\ \lambda$. All reported quantities come from the final iterate; the best-so-far E is used only for the convergence plot.

Convergence and verification at λ = 0.1

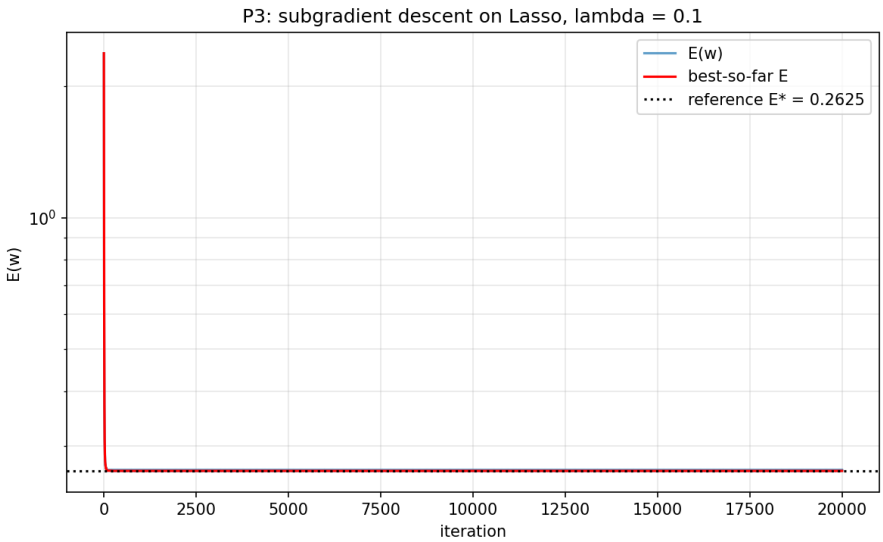


Figure 8. E(w) and best-so-far E for subgradient descent at λ = 0.1; the dotted line is the reference E^*.

0. SETUP

```
basis: shifted Legendre, degree 9  (np.polynomial.legendre.legvander(2x-1, 9))
cond(Phi^T Phi) = 6.112e+02   (assignment says about 6e2)
lambda_max(Phi^T Phi) = 12.399;  step size eta = 0.05 (smooth-part stability needs eta < 0.1613)
20000 iterations per lambda, 21 lambdas from 1e-04 to 1e+00
"approximately zero" means |w_j| < 2*eta*lambda
```

1. CONVERGENCE AND VERIFICATION AT λ = 0.1

```
E at iteration 0 / 100 / 1000 / 20000 = 2.3807 / 0.2638 / 0.2637 / 0.263570
final E(w)          = 0.263570
best-so-far E       = 0.262619   (reference E* = 0.2625)
E is not monotone: it rose on 9625 of 20000 steps

(b) optimality conditions on the final iterate
  j        w_j    Phi_j^T r  lam*sign(w_j)   status
  0    -0.0059     +0.0431        -0.1000   ~0   (need |Phi_j^T r| <= lam)
  1    -0.9296     -0.0940        -0.1000   non-zero (need Phi_j^T r = lam*sign)
  2    +0.0431     +0.1133        +0.1000   non-zero (need Phi_j^T r = lam*sign)
  3    +1.1147     +0.1054        +0.1000   non-zero (need Phi_j^T r = lam*sign)
  4    +0.0303     +0.1144        +0.1000   non-zero (need Phi_j^T r = lam*sign)
  5    -0.2025     -0.0950        -0.1000   non-zero (need Phi_j^T r = lam*sign)
```

```
  6    -0.0016     +0.0683        -0.1000   ~0   (need |Phi_j^T r| <= lam)
  7    -0.0251     -0.0952        -0.1000   non-zero (need Phi_j^T r = lam*sign)
  8    +0.0018     +0.0560        +0.1000   ~0   (need |Phi_j^T r| <= lam)
  9    +0.0005     +0.0529        +0.1000   ~0   (need |Phi_j^T r| <= lam)
max |Phi_j^T r - lam sign(w_j)| over non-zero coefficients = 0.0144
max |Phi_j^T r| / lam over ~0 coefficients               = 0.6830   (should be <= 1)

(c) comparison with the reference solution
  j   w (ours)   w* (ref)       diff
  0    -0.0059     +0.000    -0.0059
  1    -0.9296     -0.928    -0.0016
  2    +0.0431     +0.045    -0.0019
  3    +1.1147     +1.117    -0.0023
  4    +0.0303     +0.032    -0.0017
  5    -0.2025     -0.201    -0.0015
  6    -0.0016     +0.000    -0.0016
  7    -0.0251     -0.026    +0.0009
  8    +0.0018     +0.000    +0.0018
  9    +0.0005     +0.000    +0.0005
||w - w*||_2 = 0.0076   (chatter of order eta*lambda = 0.005000000000000001 is expected)
```

RMSE sweep and coefficient path

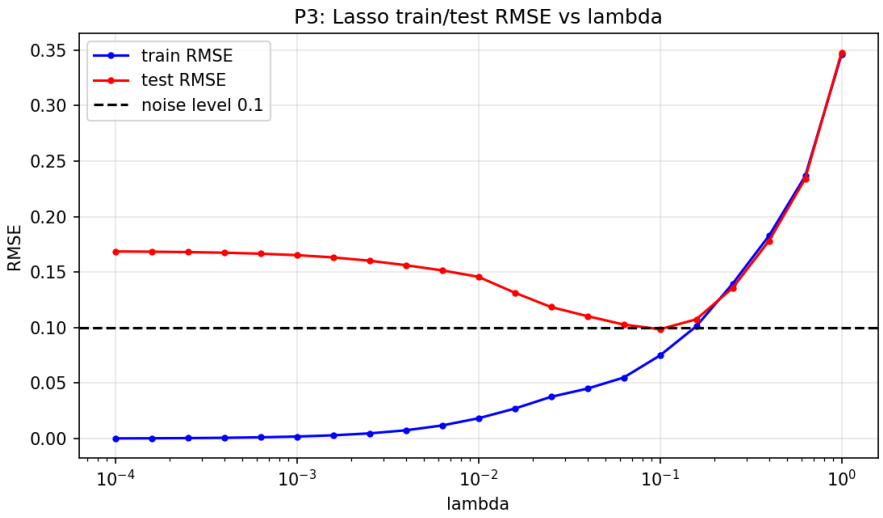


Figure 9. Lasso train and test RMSE vs λ; the dashed line is the noise level 0.1.

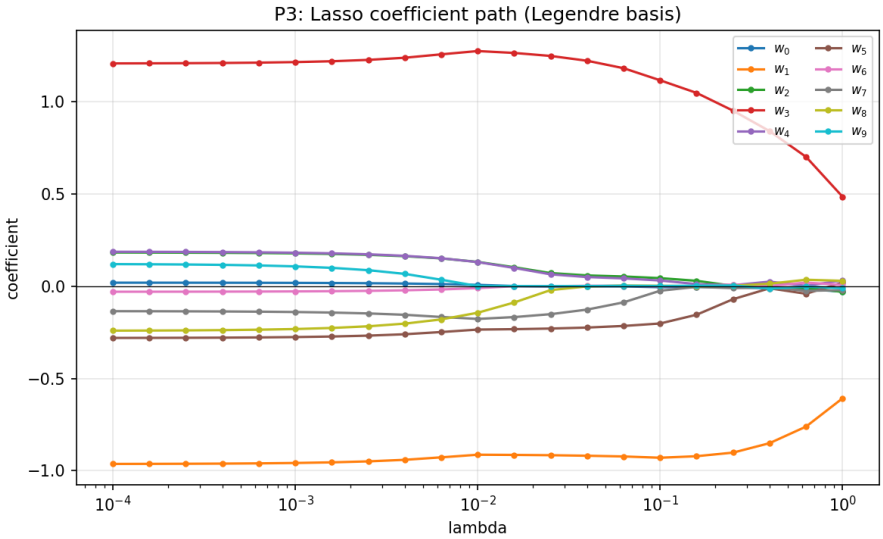


Figure 10. Lasso coefficient path: each w_j vs λ.

2-3. SWEEP OVER λ (all numbers from the final iterate)

```
    lambda  train RMSE   test RMSE   ||w||_1  #~0   zero indices
    0.0001      0.0002      0.1687    3.3628    0   []
    0.0002      0.0003      0.1684    3.3607    0   []
    0.0003      0.0005      0.1681    3.3575    0   []
    0.0004      0.0007      0.1675    3.3523    0   []
    0.0006      0.0012      0.1667    3.3441    0   []
    0.0010      0.0019      0.1653    3.3310    0   []
    0.0016      0.0030      0.1633    3.3104    0   []
    0.0025      0.0047      0.1603    3.2777    0   []
    0.0040      0.0075      0.1563    3.2258    0   []
    0.0063      0.0119      0.1516    3.1437    0   []
    0.0100      0.0183      0.1458    3.0228    1   [9]
    0.0158      0.0270      0.1315    2.8684    3   [0, 6, 9]
    0.0251      0.0376      0.1186    2.7028    3   [0, 6, 9]
    0.0398      0.0450      0.1104    2.6027    4   [0, 6, 8, 9]
    0.0631      0.0550      0.1028    2.5072    4   [0, 6, 8, 9]
    0.1000      0.0749      0.0985    2.3552    4   [0, 6, 8, 9]
    0.1585      0.1012      0.1075    2.1753    6   [0, 4, 6, 7, 8, 9]
    0.2512      0.1395      0.1358    1.9605    7   [0, 2, 4, 6, 7, 8, 9]
    0.3981      0.1827      0.1781    1.7827    8   [0, 2, 4, 5, 6, 7, 8, 9]
    0.6310      0.2366      0.2342    1.6114    8   [0, 2, 4, 5, 6, 7, 8, 9]
    1.0000      0.3461      0.3478    1.2718    8   [0, 2, 4, 5, 6, 7, 8, 9]
```

4. SELECTING λ^*

```
lambda* = argmin test RMSE = 1.0000e-01
test RMSE at lambda*       = 0.0985   (train RMSE 0.0749, noise level 0.1)
number of ~0 coefficients  = 4   indices = [0, 6, 8, 9]
coefficients at lambda*:
   w[0] = -0.0059   (~0)
   w[1] = -0.9296
   w[2] = +0.0431
   w[3] = +1.1147
   w[4] = +0.0303
   w[5] = -0.2025
   w[6] = -0.0016   (~0)
   w[7] = -0.0251
   w[8] = +0.0018   (~0)
   w[9] = +0.0005   (~0)
```

Lasso vs Ridge

5. LASSO vs RIDGE (both in the Legendre basis, same λ grid)

```
Ridge best lambda = 1.5849e-01 (test RMSE 0.1422)
Lasso best lambda = 1.0000e-01 (test RMSE 0.0985)

method    best lambda  test RMSE   ||w||_2  #|w_j|<0.01
Lasso       1.000e-01     0.0985    1.4667            4
Ridge       1.585e-01     0.1422    1.4406            1

Ridge coefficients at its best lambda:
   w[0] = +0.0032
   w[1] = -0.9604
   w[2] = +0.1084
   w[3] = +1.0043
   w[4] = +0.1064
   w[5] = -0.2627
   w[6] = -0.0138
   w[7] = -0.0446
   w[8] = -0.0941
   w[9] = +0.2032
```

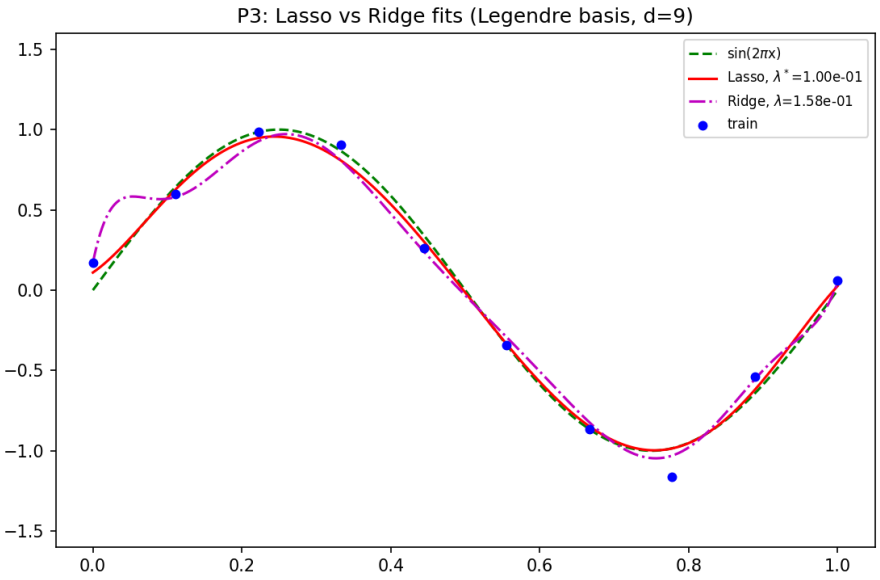


Figure 11. Lasso fit at λ^*, Ridge fit at its own best λ, training data and $\sin(2\pi x)$.

Discussion

L1 gives exact zeros because of the optimality condition: a coefficient can sit at zero only while the correlation of its feature with the residual stays within λ ($|\Phi_j^\top r| \le \lambda$), and the corners of the l1 ball (diamond) lie on the axes, where the SSE contours touch first. The coefficient path shows the coefficients shrinking and then switching off one after another (0 of 10 are ≈ 0 at $\lambda = 10^{-4}$, 8 of 10 at $\lambda = 1$). Ridge's circular l2 ball is touched away from the axes, so it shrinks every coefficient without setting any to zero: at its best λ 1 of 10 Ridge coefficients have $|w_j| < 0.01$, against 4 of 10 for Lasso. Past $\lambda^* = 10^{-1}$, the penalty starts to dominate: train RMSE rises monotonically ($0.075 \to 0.346$ at $\lambda = 1$) and test RMSE rises too ($0.099 \to 0.348$), i.e. the model underfits.

P4

Bayesian Regression — MAP/Ridge Equivalence and Effect of N

Method

Fixed hyperparameters: noise variance 0.01 ($\beta = 100$), prior precision $\alpha = 0.05$, polynomial degree 3. The posterior over the weights is $\mathrm{N}(m_N, S_N)$ with $S_N^{-1} = \alpha\,\mathrm{I} + \beta\,\Phi^\top\,\Phi$ and $m_N = \beta\,S_N\,\Phi^\top\,\mathrm{t}$. The predictive distribution at x is $\mathrm{N}(\mathrm{m(x)}, s^2(x))$ with $s^2(x) = 1/\beta + \phi(x)^\top\,S_N\,\phi(\mathrm{x})$. The MAP/ridge estimate with $\lambda = \alpha/\beta$ is obtained by solving the ridge linear system.

Predictive distribution and MAP equivalence (N = 10)

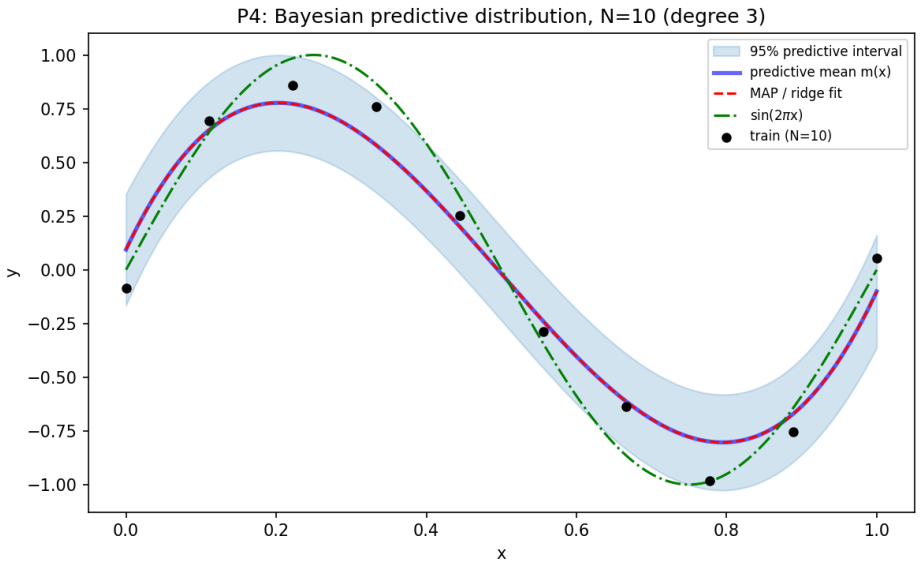


Figure 12. Predictive mean with the 95% interval, the MAP/ridge curve (dashed), the data and $\sin(2\pi x)$.

1. POSTERIOR AND PREDICTIVE DISTRIBUTION (`p4_train_N10.csv`, N = 10)

```
posterior mean m_N and posterior std of each weight:
   w[0]: m_N =   +0.0938   std =   0.0868
   w[1]: m_N =   +7.3679   std =   0.7349
   w[2]: m_N =  -22.8093   std =   1.7431
   w[3]: m_N =  +15.2470   std =   1.1481

predictive std s(x):  min = 0.1108 (at x = 0.487),  max = 0.1338 (at x = 1.000)
noise-only std beta^-1/2 = 0.1000
95% interval width: min = 0.4343, max = 0.5244
```

2. MAP / RIDGE EQUIVALENCE CHECK

```
  j        w_MAP          m_N   difference
  0    +0.093837    +0.093837    -4.74e-14
  1    +7.367889    +7.367889    +8.59e-13
  2   -22.809349   -22.809349    -2.33e-12
  3   +15.247000   +15.247000    +1.58e-12

||w_MAP - m_N||_2             = 2.943e-12
max |y_MAP(x) - m(x)| on grid = 5.862e-14
=> the MAP/ridge curve and the Bayesian predictive mean coincide to rounding error.
```

Effect of N

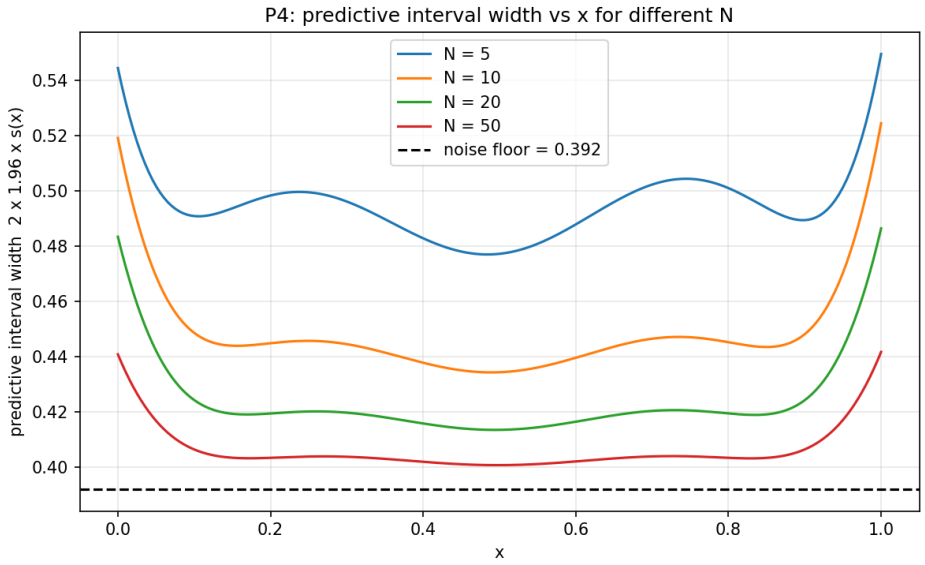


Figure 13. Predictive interval width vs x for N = 5, 10, 20, 50; the dashed line is the irreducible-noise floor.

3. EFFECT OF N (α, β fixed)

```
   N  mean width  max width  mean phi^T S phi  max phi^T S phi  mean width / floor
   5      0.4952     0.5494           0.00597          0.00964               1.263
  10      0.4491     0.5244           0.00314          0.00789               1.146
  20      0.4242     0.4863           0.00172          0.00539               1.082
  50      0.4064     0.4417           0.00075          0.00269               1.037

noise floor (irreducible) = 0.3920  (beta^-1 = 0.01 never shrinks with N)
```

Discussion

More data raise the posterior precision $S_N^{-1} = \alpha\,\mathrm{I} + \beta\,\Phi^\top\,\Phi$ (each new point adds $\beta\,\phi\,\phi^\top$), so the posterior covariance S_N shrinks and the parameter-uncertainty term $\phi(x)^\top\,S_N\,\phi(\mathrm{x})$ falls: the mean of that term drops from 0.0060 at N = 5 to 0.00075 at N = 50, and the interval narrows from an average width of 0.495 to 0.406. For N = 10 the band is narrowest near x = 0.49 and widest at x = 1.00. The noise term $\beta^{-1} = 0.01$ is a property of the observations, not of the weights, so it does not depend on N: the width can only approach the floor 0.392 and never go below it, however much data we collect.